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Seminar Report
On
Football Player Tracking Using Computer Vision Techniques
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CERTIFICATE

This is to certify that Mr. Aaditya Sharma of T.Y. B.Tech., School of Computer Engineering & Technology, Department of Computer Engineering & Technology, CSE-Core, Semester – VI, PRN. No. 1032231950, has successfully completed seminar on

Football Player Tracking Using Computer Vision Techniques

To my satisfaction and submitted the same during the academic year 2025- 2026 towards the partial fulfillment of degree of Bachelor of Technology in School of Computer Engineering & Technology under Dr. Vishwanath Karad MIT- World Peace University, Pune.

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LIST OF ABBREVIATIONS

Table 1: Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CV	Computer Vision
DNN	Deep Neural Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
HOG	Histogram of Oriented Gradients
IoU	Intersection over Union
KF	Kalman Filter
mAP	Mean Average Precision
MOT	Multi-Object Tracking
NMS	Non-Maximum Suppression
ReLU	Rectified Linear Unit
ResNet	Residual Neural Network
ROI	Region of Interest
RPN	Region Proposal Network
SVM	Support Vector Machine
SORT	Simple Online and Realtime Tracking
STP	Spatiotemporal Processing
VOD	Video Object Detection
YOLO	You Only Look Once

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ABSTRACT

The rapid advancement of computer vision and deep learning technologies has significantly transformed the domain of sports analytics, particularly in football.

Traditional methods of performance analysis relied heavily on manual observation, which was time-consuming, error-prone, and limited in scalability. With the emergence of automated systems, it has become possible to analyse large volumes of match data efficiently, providing valuable insights into player performance, team strategies, and game dynamics.

Tracking players in a football match using video analysis is challenging.

Players move quickly, often block each other from the camera's view, and those on the same team look visually similar due to identical jerseys. This report presents a computer vision-based system that automatically detects and tracks all players in real time using a nine-step pipeline.

The system uses four main algorithms: YOLO for detecting players in each frame, a Convolutional Neural Network (CNN) which identifies individual players based on their appearance, DeepSORT to keep track of each player across multiple frames, and a Kalman Filter to predict where a player moves next. Together, these tools allow the system to follow every player throughout a match, even if they are briefly untraceable or clustered together.

Overall, this report emphasizes the importance of developing robust, scalable, and efficient systems for football analytics. Future research should focus on improving occlusion handling, reducing computational complexity, enhancing dataset quality, and bridging the gap between theoretical research and practical applications.

Keywords: Football player tracking, computer vision, YOLO, Convolutional Neural Network, DeepSORT, Kalman Filter, multi-object tracking, sports analytics, object detection, deep learning, re-identification, spatiotemporal analysis.

1. INTRODUCTION

Traditional tracking solutions relied on GPS sensors or wearable devices, which are costly and cannot always be used in official competitions. As a result, many teams and analysts lack access to accurate player movement data.

Another major challenge is ensuring accurate refereeing decisions, especially for situations such as offside and foul detection. Human referees may struggle to make these decisions due to the speed and complexity of the game.

Computer vision and deep learning techniques can automatically detect and track players from match videos, providing a cost-effective and scalable alternative.

This method of tracking enables player identification, assigns identities to them, and tracks their movements across video frames. Some of the methods used here include object detection, feature extraction, and multi-object tracking. There have been significant advancements in deep learning models, particularly in CNNs, that have improved detection capabilities.

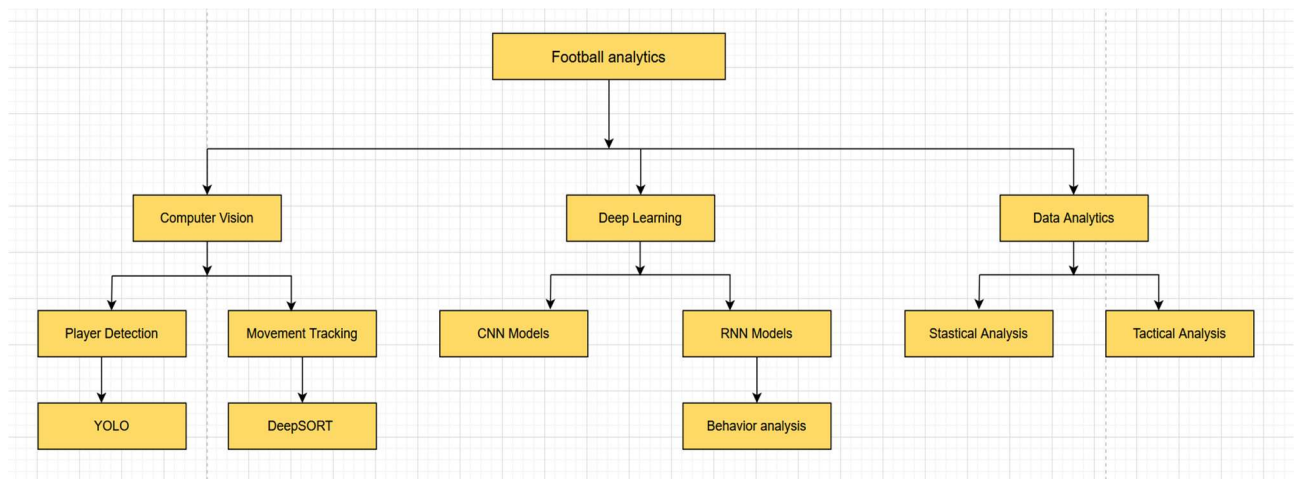


Figure 1: Different algorithms and techniques for football analytics

Player tracking technology is important for decision-making, especially since it provides spatial location of players. Foul detection and offside detection are two cases when it becomes necessary to track movement of players relative to their opponents and teammates.

The technical content of this report is organized in three main parts. Part one will be the introduction of the notion and significance of tracking football players through computer vision. Part two will include a thorough review of thirteen relevant papers along with their methods, benefits, and drawbacks. Lastly, part three will describe the technology behind contemporary tracking systems.

2. LITERATURE SURVEY

Many research projects have looked at tracking football players using computer vision methods. These investigations especially concentrate on raising detection precision, tracking uniformity, and computational efficiency.

Early studies like “A survey on player tracking in soccer videos” showed that player tracking consists in several steps including player detection, labeling, and occlusion management. It underlined that fast motion, lighting changes, and occlusion are the key difficulties influencing system effectiveness.

Another computer vision applications in sports research examined the increasing relevance of vision-based systems in supplanting sensor-based monitoring methods. It emphasized how computer vision could help with coaching, broadcasting, and referee decision-making.

Recent research has concentrated on methods driven by deep learning. Deep neural networks were found to be able to precisely identify and monitor players from video footage in research on amateur football analytics. These techniques use tracking algorithms together with object detection models like YOLO.

Likewise, research employing YOLO-based identification and DeepSORT tracking have demonstrated enhanced capability in multi-object tracking applications. Real-time identification and tracking of several players concurrently are made possible by these techniques.

Studies based on IEEE further investigated sophisticated tracking systems combining motion prediction and data association methods with detection. Although these approaches enhance identity consistency across frames, they still run into difficulties in busy environments.

Appearance-based tracking models, which utilize visual cues like jersey color and texture to keep player identification during occlusion, are another significant contribution. But these models struggle when the players look alike.

Motion analysis and spatio-temporal data research has helped us to better grasp player behavior and strategic patterns. These methods examine player paths over time to find useful data including movement patterns and team formations.

Table 2: Literature review

Ref	Method / Technique Used	Advantages	Disadvantages	Key Contribution	Scope of Improvement
[1]	Deep Learning (CNN-based analytics)	High accuracy, detailed insights	High computational cost, requires large data	Football performance analytics	Improve real-time efficiency
[2]	Computer Vision vs Motion Capture	Reliable tracking accuracy, realistic evaluation	Limited scalability, expensive setup	Validation of vision systems	Reduce cost and improve scalability
[3]	Deep Learning Player Tracking	Robust detection, good feature learning	Sensitive to motion blur, affected by lighting	Improved tracking accuracy	Handle dynamic environments better
[4]	Automated CV Tracking	Fully automated system, minimal manual effort	Occlusion challenges, identity switching	End-to-end tracking system	Better occlusion handling
[5]	YOLO-based Detection	Fast detection, suitable for real-time use	Accuracy declines in crowded scenes, misses small objects	Real-time tracking	Better detection in dense scenes

[6]	Deep Neural Network Tracking	High tracking consistency, improved accuracy	High resource usage, slower processing	Multi-object tracking improvement	Optimize computation
[7]	Appearance based Tracking	Handles occlusion well, maintains identity	Similar jersey problem, feature confusion	Identity preservation	Enhance re-identification
[8]	Scalable Deep Learning Video Analysis	Supports large-scale analysis, useful for sports analytics	Requires large datasets, training complexity	Large-scale analysis capability	Minimize data dependency
[9]	YOLO and DeepSORT	Real-time performance, efficient tracking	Identity switching, struggles in dense scenes	Efficient tracking pipeline	Improve identity stability
[10]	SoccerNet Dataset	Benchmark dataset, supports research	Limited annotations, dataset imbalance	Standard dataset for evaluation	Expand annotations
[11]	Space-Time Analysis	Tactical insights, movement understanding	Complex modeling, high computation	Game strategy understanding	Simplify models
[12]	Survey-Based Review	Comprehensive coverage, identifies challenges	Lack of implementation, general overview	Research overview & applications	Apply in real-world systems

[13]	DL-based Behaviour Analysis	Action recognition, behaviour insights	High computational complexity, needs large data	Player behaviour analysis	Improve efficiency and scalability
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3. DETAILS OF DESIGN AND TECHNOLOGY

Current football player tracking systems use a multi-stage, structured pipeline.

The first step in the process is video acquisition, which uses tactical or broadcast cameras as input. Field segmentation, noise reduction, and frame extraction are examples of preprocessing procedures. Deep learning models like YOLO, Faster R-CNN, and SSD are used to detect players. Bounding boxes are created around players by these models. YOLO's quickness and efficiency make it popular.

After that, feature extraction is carried out to record player attributes like texture and color. These characteristics are utilized to differentiate between players. Players are given unique IDs by multi-object tracking systems like DeepSORT. To preserve identity, these systems employ feature matching and motion prediction. Motion estimation is done using Kalman filters.

Temporal consistency and appearance-based models are used to handle occlusion. It is still a significant obstacle. Player mobility throughout time is examined using spatiotemporal analysis.

This covers interactions, speed, and trajectory. Temporal consistency and appearance-based models are used to handle occlusion. It is still a significant obstacle, though.

Player mobility throughout time is examined using spatiotemporal analysis. This includes interactions, speed, and trajectory analysis. Applications like offside and foul detection are included in event detection. These need precise timing and location information. Player trajectories, heatmaps, and performance indicators are included in the finished product.

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3.1 Current System Architectures

These methodologies for tracking football players are characterized by uncomplicated processing sequences. Footage of a match is captured by a camera, from which individual frames are extracted. Subsequently, conventional image processing algorithms, such as background subtraction and edge detection, are employed to distinguish moving entities.

Player identification is then achieved through the utilization of manually designed feature descriptors, like Histogram of Oriented Gradients (HOG), in conjunction with Support Vector Machines (SVM) for classification purposes. The tracking of players across consecutive frames is managed by rudimentary centroid-based techniques or basic Kalman Filter implementations that exclusively utilize positional information.

These systems operate on a sequential paradigm: video input, frame extraction, foreground object identification, the generation of bounding boxes, and the recording of player positions.

Although these systems proved to be effective under controlled circumstances, their underlying architecture is inflexible. Each component was developed based on predetermined assumptions, including consistent illumination, unobstructed camera perspectives, and minimal player intervisibility. Any deviation from these assumptions during a match would result in a significant degradation of performance.

Furthermore, a notable architectural drawback is the reliance on databases. These systems need the manual annotation of training data for each novel stadium, camera setup, or lighting environment. This approach lacks generalizability. A model trained on footage captured during daylight hours would exhibit sub-optimal performance under artificial lighting, and a model trained with a single stationary camera would encounter difficulties with broadcast feeds that involved camera panning and zooming.

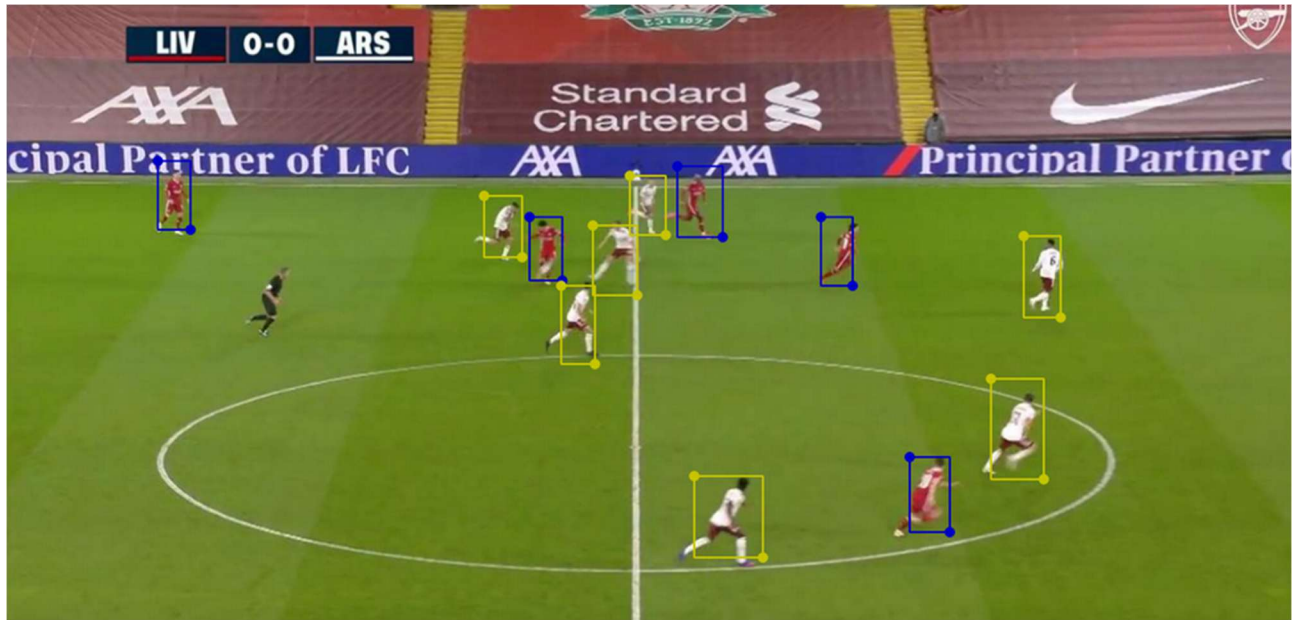


Figure 2: An image showing player detection

3.1.1 Limitations of Current Systems

An examination of the shortcomings of existing systems directly informed the design decisions implemented in the proposed methodology:

Occlusion: In situations where players overlap, such as during aerial duels, tackles, or set-piece plays, preceding systems are unable to maintain continuous tracking. There is no inherent capability to re-establish a player's identity once they become obscured by another.

Identity switching: Within densely populated areas of the field, tracking algorithms frequently reassign player identities incorrectly.

Motion Blur: Fast movement like sprints, shots, sudden direction changes causes blurring at standard frame rates, degrading the quality of both detection and feature extraction.

The computational demands of deep learning inference, even solely for detection, necessitate substantial GPU capacity, rendering the real-time execution of comprehensive tracking pipelines on standard hardware a considerable hurdle for numerous systems.

Visual homogeneity: characterized by identical team jerseys, complicates the differentiation of players by tracking systems. Conventional and even some deep learning-based feature extraction techniques struggle to maintain player identity consistently, particularly when viewed from a distance.

Dependency on datasets: the development of dependable models is contingent upon extensive and meticulously annotated datasets, the creation of which is both costly and time-intensive, thereby constraining the scalability and adaptability of tracking systems across diverse operational contexts.

3.2 Algorithms Used

3.2.1 YOLO (You Only Look Once)

General Functionality: YOLO is a real-time algorithm for object detection. In contrast to conventional detection methods that repeatedly scan an image via region proposals, YOLO processes the entire image in a single forward pass through a neural network. It partitions the image into a grid, with bounding boxes and class probabilities. This approach yields significantly greater speed compared to two-stage detectors, while maintaining comparable accuracy.

Two-stage detectors, such as Faster R-CNN, offer superior accuracy but are too computationally intensive for real-time applications. While SSD is faster, it experiences a reduction in accuracy when detecting small or partially obscured objects. YOLO achieves the requisite balance for live football analysis, prioritizing speed without a substantial compromise in precision. Subsequent anchor-free iterations (YOLOv8 and later) further enhance the detection of diminutive players in wide-angle imagery.

System Application: YOLO is utilized to process each incoming video frame, creating bounding boxes around every identified player. Its operational speed is sufficient to maintain synchronization with 30 frames per second match footage, thereby serving as the foundational detection component for the entire processing pipeline.

3.2.2 Convolutional Neural Network (CNN)

General Functionality: A CNN is a deep learning architecture engineered for processing data organized in a grid, most commonly images. It employs layers of trainable filters that traverse an image, progressively extracting features. These features range from basic edges and textures in the initial layers to intricate shapes and patterns in deeper layers. CNNs autonomously acquire these features from data, removing the need for manually engineered descriptors.

Traditional descriptors, such as Histogram of Oriented Gradients (HOG), encode static patterns that are not adaptable to novel conditions. CNNs, conversely, learn representations specifically optimized for the visual characteristics of the training data.

In football, this allows the system to distinguish between two players wearing the same jersey by recognizing subtle variations in their physique, stance, and even their jersey number, differences that a manually created description wouldn't easily pick up.

When YOLO identifies a player and outlines them with a bounding box, the CNN analyzes that captured image and generates a concise feature vector, essentially a numerical signature of the player's appearance. This vector is then used by the tracking component to identify the same player in later frames, even if they momentarily vanish from view.

3.2.3 DeepSORT (Deep Simple Online and Realtime Tracking)

General functionality:

DeepSORT is an algorithm for tracking multiple objects, assigning them unique identifiers and keeping track of those identities frame after frame. At each moment, it tackles a data pairing challenge: figuring out which new detections match up with existing tracked objects. It achieves this by considering two things at once: how far apart the predicted and actual positions are, and how visually alike the appearance descriptions are. The Hungarian algorithm is employed to find the best possible match.

Less complex trackers, like SORT, rely only on location data for matching. In a sport like soccer, where players are constantly crossing paths, just using location isn't enough as two

players passing each other would end up having their IDs switched. DeepSORT's use of appearance descriptions offers a second, separate piece of information that stops this from happening. No tracking method that relies solely on movement can maintain the same level of consistent identity in busy, fast-moving situations.

Application in this system:

DeepSORT gets the appearance vectors created by the CNN along with the bounding boxes from YOLO. It keeps a collection of embeddings for every active track and compares new detections to this collection to make decisions about assignments. Players retain their assigned IDs across different frames, camera movements, and short periods of being hidden – something that a tracking system relying only on position-based information couldn't achieve.

3.2.4 Kalman Filter

What it does overall:

Basically, the Kalman Filter is an algorithm used for figuring out where something is when you're not totally sure. You give it a bunch of messy readings, and it spits out the best guess of what's really going on. It does this by comparing a guess based on how you think things move with the new readings you get. It keeps track of two things: your best guess of the situation and how confident you are in that guess, updating both as time goes on.

Why it's not substituted:

Just connecting the dots between where you last saw something and where you see it now is too simple. It doesn't handle the confusion from losing track of something or getting fuzzy location data. Other fancy methods, like particle filters, are much more demanding on your computer and don't really do a better job for something as predictable as how football players move. The Kalman Filter is light on resources, makes good mathematical sense, and works great for the mostly straightforward movements of players over short periods.

Use in our application:

The Kalman Filter is working behind the scenes in DeepSORT for every player we're tracking. When we spot a player, the filter gets updated with their location. If a player suddenly disappears for a bit, maybe they're behind someone else, or the detection system missed them then the filter uses its movement prediction to guess where they probably are. This guessed spot is used to keep a placeholder for that player and look for them in the next picture, stopping us from losing track of them when they're hidden for a short while.

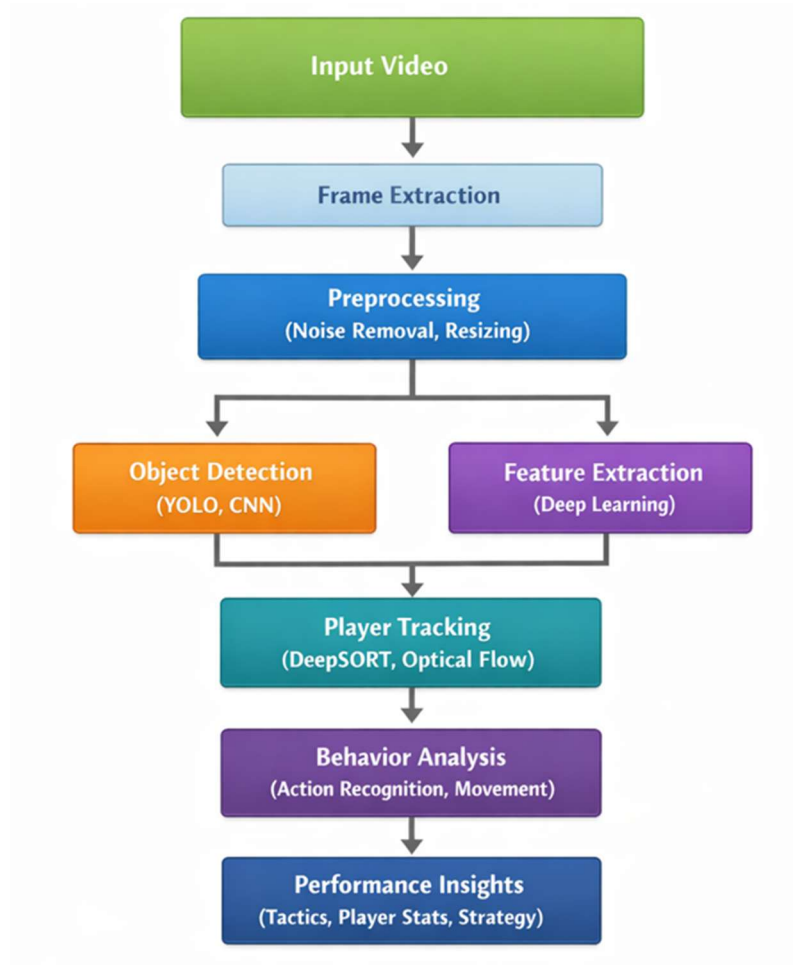


Figure 3: System Architecture

3.3 Proposed System

The proposed system is designed to address the limitations described in Section 3.2 by integrating the four algorithms above into a unified, nine-stage pipeline.

Stage 1 — Video Acquisition: Match footage is captured via broadcast or tactical cameras. The system supports both live streams and pre-recorded video.

Stage 2 — Preprocessing: Frames sizes are changed and normalized. Background subtraction removes static pitch regions to reduce false detections. Noise reduction handles compression artifacts from broadcast encoding.

Stage 3 — Player Detection: YOLO processes each frame and generates bounding boxes around all detected players. A confidence threshold filters out weak predictions, and Non-Maximum Suppression removes duplicate boxes for the same player.

Stage 4 — Feature Extraction: Each player's image within a bounding box goes through a Convolutional Neural Network (CNN) to create a 512-dimensional "appearance embedding." This is basically a condensed way to describe what the player looks like at that specific time.

Stage 5 — Multi-Object Tracking: The DeepSORT algorithm takes these bounding boxes and embeddings. It figures out which new player sightings match up with existing players by looking at how similar they look and where they're predicted to be, making sure players keep the same ID from one video frame to the next.

Stage 6 — Occlusion Handling: If a player gets completely hidden from view, the Kalman Filter keeps track of where they're likely to be for up to 30 frames. When the player shows up again, their appearance embedding is used to confirm it's really them before giving them back their original ID. This really helps fix the problems of players getting the wrong ID or losing track of them that other systems struggle with.

Stage 7 — Spatiotemporal Analysis: The paths players take, which are initially in pixel terms, are converted into real-world field coordinates using a homography transformation. Then, things like speed, how far they've run, and which way they're moving are calculated for each player.

Stage 8 — Event Detection: Important game moments like pressing plays, tackles, and set pieces are identified. This is done by combining rules that check when players are close to each other with a simple classifier that learns from tagged event data over time.

Stage 9 — Output Visualisation: The system generates a video with player IDs and their movements shown on top, a bird's-eye view of the field showing tactics, and a dashboard after the game with heatmaps and stats.

This new design is better than older ones because it uses CNNs to learn features instead of hand-coding them, adds appearance-based re-identification to handle players being hidden, and uses the Kalman Filter to fill in the gaps when players aren't detected. By doing this, it fixes the three main issues that caused problems in earlier systems.



Figure 4: Different layers of the system

4. CONCLUSION

This project offers a thorough way to track football players using computer vision, tackling the issues with older and current systems. It uses cutting-edge deep learning models like YOLO for spotting players, Convolutional Neural Networks for pulling out important details, and DeepSORT for following multiple objects. This setup allows for precise and steady tracking of players during a game.

The system is built in a step-by-step process, making it efficient to go from getting video footage to spotting specific events like foul detection.

A significant advantage of this project is how it can be used in real games like automatically spotting offsides and fouls. By accurately figuring out where players are and how they interact, the system can help referees, coaches, and analysts make better calls. This really shows how valuable computer vision is for analyzing football today.

Even with these improvements, there are still some hurdles, such as the high computing power needed, the reliance on big datasets, and performance issues in tricky situations. Future work could concentrate on making the models run faster in real-time, improving how players are identified again, and using footage from multiple cameras for even better accuracy.

Overall, this system offers a flexible, efficient, and affordable way to track football players, pushing forward smart sports analysis and automated decision-making tools.

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